USES OF LASERS

Medical
Lasers are used for extremely precise cutting during surgery, cauterizing wounds, and for corrective eye surgery.

Surveying
Cheap, low-power lasers produce thin beams in straight lines, useful for builders and surveyors.

Welding
Some lasers can be automated for high-speed work, such as joining parts of car bodies, pots, and pans.

Manufacture
Lasers are used for precise cutting of fabrics for the clothing industry and etching out letters or numbers on keyboards.

Entertainment
Lasers provide light shows at concerts, often drawing intricate patterns in theatrical smoke. CD and DVD players also use lasers.

Telecommunications
Infrared laser diodes send digital information along optical fibers across worldwide networks.

GAS LASERS

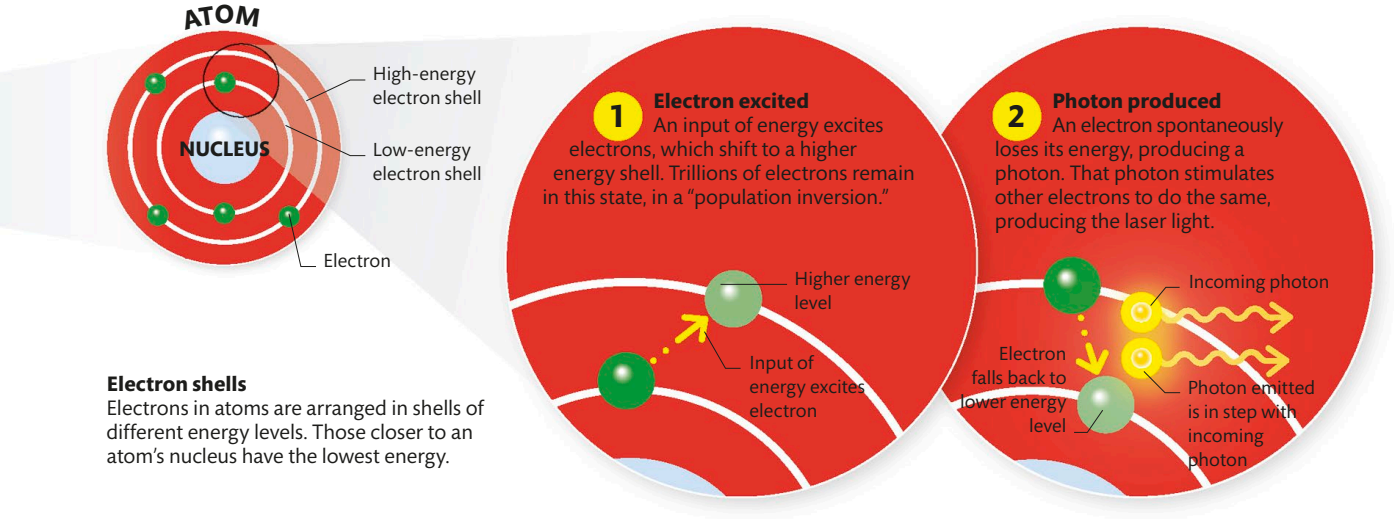
Not all lasers are solid-state, semiconductor laser diodes. Many of the most powerful are gas lasers, in which the excited electrons are in atoms of a gas. Lasers with carbon dioxide gas as the lasing medium are used to cut and weld car parts, for example.

How photons are produced

The photons (particles of light) that make up a laser beam are created by a process called stimulated emission. They are produced by electrons in atoms in the lasing medium—in a solid-state laser diode, this is the undoped semiconductor in the semiconductor sandwich (see opposite). An electric current (or, in some lasers, a burst of light) excites (boosts) electrons to a higher energy level. When an electron falls back to the lower energy level, the extra energy is released as a photon. The photon travels through the lasing medium, stimulating more excited electrons to release photons. The color of the laser light depends on the difference in energy between the higher and lower levels.

LASER BEAM

LASERS CAN MEASURE THE DISTANCE FROM EARTH TO THE MOON TO WITHIN A FEW CENTIMETERS



Electron shells
Electrons in atoms are arranged in shells of different energy levels. Those closer to an atom's nucleus have the lowest energy.